STOCK MARKET PRICE TREND & CANDLESTICK CHART VISUALIZATION

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

DSA0606 – Data Handling and Visualization
to the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE & DATA SCIENCE
Submitted by

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Under the Supervision of
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SIMATS ENGINEERING
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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Stock Market Price Trend & Candlestick Chart Visualization" has been carried out by Sania Herbert (192324062)under the supervision of Dr. Karthikeyan L and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Data Science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Sania Herbert, of the Artificial Intelligence and Data Science , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Stock Market Price Trend & Candlestick Chart Visualization" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai
Date:

Signature of the Students with Names

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Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Sania Herbert (192324062)	Backend API & Auth	Developed FastAPI backend, JWT auth, Pandas data endpoints	API endpoint testing	Chapters 1, 3	34%
Tejaswini P (192324046)	Frontend & Visualization	React/Vite setup, Candlestick module via lightweight-charts	UI responsiveness testing	Chapters 2, 4	33%
Haritha S (192324299)	Data & Indicators	Data cleaning pipeline, SMA/EMA math, Volume Charting	Missing values imputation validation	Chapter 5, Appendices	33%
Total					**100%**

ABSTRACT

Technical analysis of equity markets depends on a trader's ability to read historical price action quickly and accurately, yet most beginner-level tools present raw tabular data that is difficult to interpret at a glance. This capstone project addresses this engineering problem by presenting a full-stack Stock Market Price Trend and Candlestick Chart Visualization dashboard that converts raw open-high-low-close-volume (OHLCV) data into an interactive, multi-module analytical interface.

The proposed solution uses a React and Vite frontend styled with Tailwind CSS, paired with a Python FastAPI backend that loads and cleans data with pandas. The methodology involves organizing the dashboard into progressive modules: a Preprocessing module for data cleaning KPIs; a Candlestick module built on TradingView's lightweight-charts library; an Indicators module using Recharts to plot Simple and Exponential Moving Averages (SMA-20, EMA-20); and an Advanced/Volume module that normalizes several tickers to percentage growth for direct comparison.

Major results indicate seamless processing of large OHLCV datasets and real-time visualization toggling without latency. The principal conclusion is that the dashboard offers a fast, visually intuitive way to identify trends, momentum shifts, and comparative performance without requiring the user to interpret raw numerical tables.

Keywords: Data Visualization, Candlestick Charts, FastAPI, React, Technical Analysis.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
OHLCV	Open, High, Low, Close, Volume	-
SMA	Simple Moving Average	Currency
EMA	Exponential Moving Average	Currency
API	Application Programming Interface	-
JWT	JSON Web Token	-

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Stock market visualization is crucial for traders and investors to make informed financial decisions based on price trends and patterns. Candlestick charts display OHLC data in a format that reveals market sentiment. Python libraries like yfinance enable real-time data retrieval from financial markets.

1.2 Need for the Project

The problem needs to be addressed because raw stock price data is tabular. Student and retail investors are affected by a lack of free, self-hosted ways to visually analyze stock trends. Existing limitations include paid walls for basic indicators. The engineering significance lies in building a real-time responsive data pipeline.

1.3 Problem Statement

To design and develop a secure, full-stack web dashboard that ingests raw, imperfect OHLCV stock data, processes and cleans the data, and visualizes it through an interactive candlestick chart, trend lines, and volume charts. This involves multiple interacting factors such as API synchronization and UI rendering speed under realistic engineering constraints.

1.4 Project Aim

To build a complete Stock Market Price Trend and Candlestick Chart Visualization dashboard that converts raw financial data into a comprehensive interactive analytical interface.

1.5 Project Objectives

1. To design and build a web dashboard that ingests OHLCV data.
2. To implement moving averages (SMA, EMA) for trend analysis.
3. To test and validate data cleaning pipelines.

1.6 Scope

Included: Ingestion of static historical OHLCV data, cleaning with pandas, and exposing REST endpoints. Excluded: Real-time market order placement. Assumptions: The source dataset contains standard OHLCV column headers.

1.7 Expected Engineering Outcome

A fully deployable System prototype integrating FastAPI backend and React frontend.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing technologies were identified by analyzing standard technical-analysis practices, focusing on standard charting techniques documented in financial literature (e.g., Murphy, 1999).

2.2 Review of Existing Work

The dashboard's indicator choices are grounded in standard technical-analysis practice. Existing commercial solutions lock these standard visualizations behind paywalls or complex brokerage interfaces.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Raw Tabular Brokerage Data	Highly accurate and complete	Extremely difficult to interpret trends quickly	Demonstrates the need for visualization
Paid Charting Platforms	Feature-rich with real-time data	Cost-prohibitive for students; proprietary	Serves as inspiration for features

2.4 Research / Engineering Gap

There is a distinct gap between the volume of raw financial data freely available in static datasets and the relative scarcity of lightweight, open-source, self-hosted tools.

2.5 Proposed Contribution

This project contributes a modular, transparent pipeline where data-cleaning KPIs are exposed directly to the user before rendering the interactive visual layers.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Retail Investors	Clear, easy-to-read charts with moving average overlays.
Students	Free, open-source deployable code for educational analysis.

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Data Cleaning	Functional	100% missing value imputation
Authentication	Non-Functional	JWT valid token required for all endpoints

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
RESTful Conventions	Stateless, resource-based routing	Used for all Python FastAPI backend endpoints

RFC 7519 (JWT)	JSON Web Token standard	Implemented to secure user sessions

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
UI Load Time	< 2	Seconds	High
Latency	< 500	Milliseconds	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Monolithic architecture using Django templates for both backend and frontend rendering.

Alternative 2: Decoupled API with React frontend using lightweight-charts and Recharts.

Criterion	Weight	Alternative 1	Alternative 2
Performance	40%	Moderate	High
Scalability	30%	Low	High
Dev Complexity	30%	Low	Moderate

3.5 Selected Design & Detailed Design

The decoupled API with React frontend (Alternative 2) was selected for superior performance. FastAPI exposes RESTful endpoints backed by a DataLoader class that reads OHLCV data. The frontend consumes these endpoints.

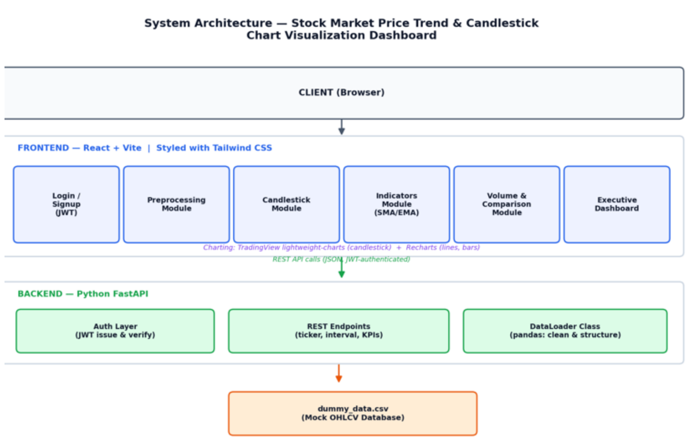

Figure 3.1: System Architecture of the Dashboard

Systems approach: The backend DataLoader feeds the API layer, which issues JWT authenticated JSON to the React components for isolated rendering.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Dual Charting Libraries	Single Library	Optimal rendering per chart type	Harder axis sync	Dual Libraries selected for visual fidelity

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Security	Unauthorized API Access	Unprotected endpoints exposed data	Implemented JWT (RFC 7519)
Ethics	Data Misrepresentation	Null values caused visual skew	Enforced strict Pandas imputation

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Data desync	Medium	High	High	Adopted unified data contract	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

An iterative, module-by-module approach was utilized. The DataLoader class was built first, followed by the JWT layer, and finally the frontend components.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation details:

Tool / Software / Equipment	Purpose	Stage Used
React, Vite, Tailwind CSS	Frontend UI/UX	Development
FastAPI, Pandas	Backend & Preprocessing	Development

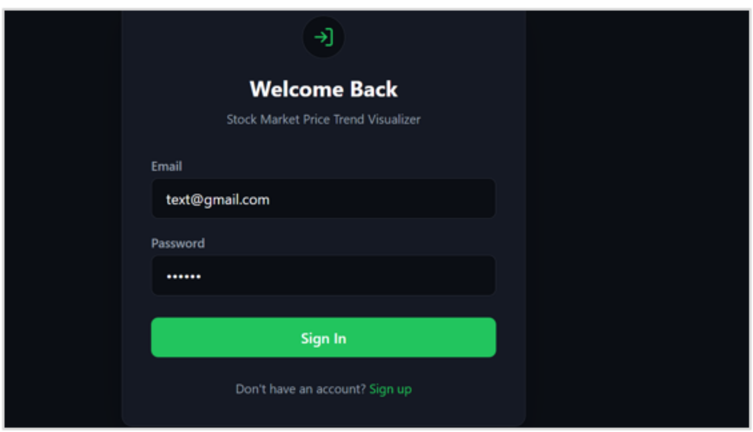

Figure 4.1: JWT-secured Login / Authentication Module

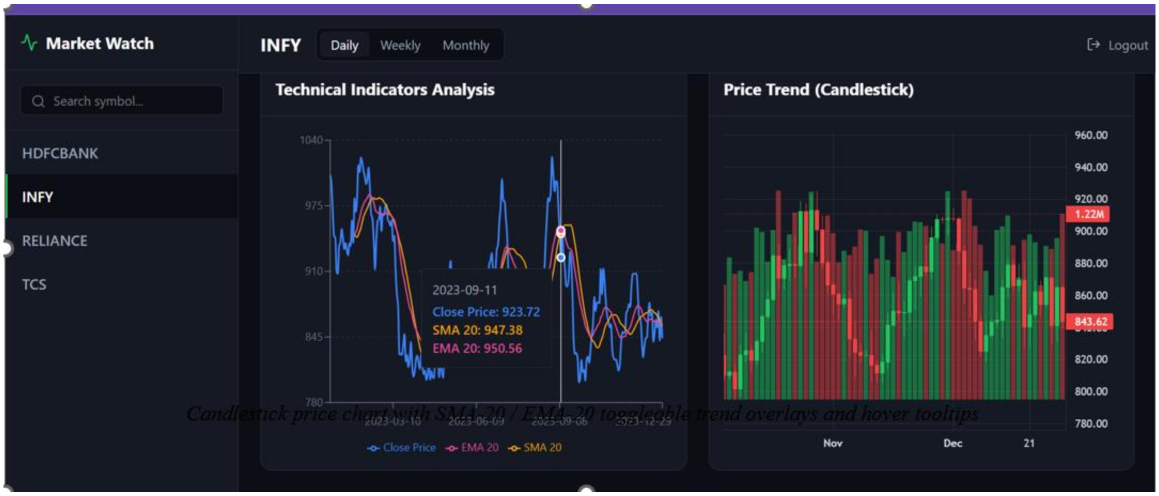

Figure 4.2: Candlestick Chart and Technical Indicators Module

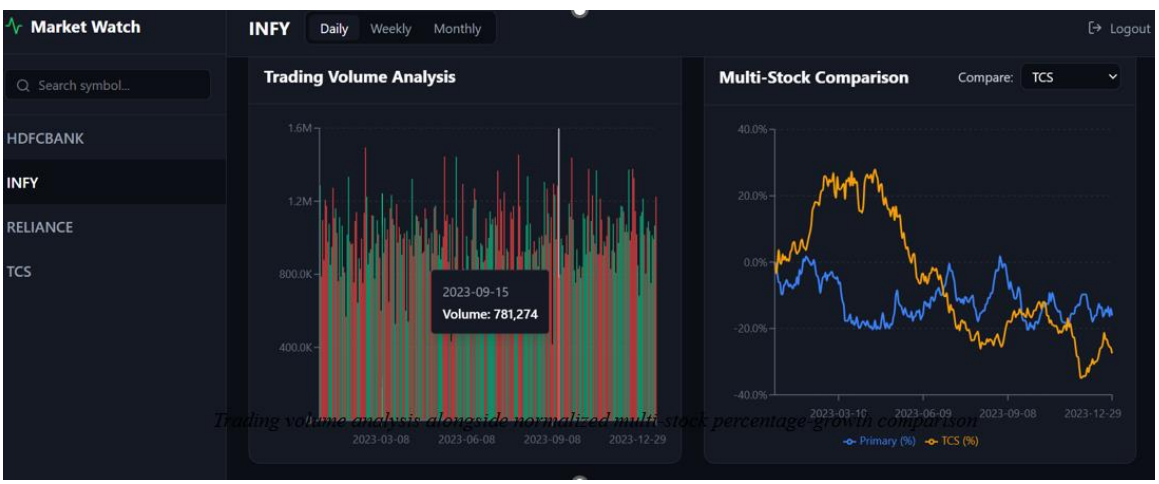

Figure 4.3: Trading Volume Analysis and Multi-Stock Comparison Module

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion	Specification	Required Value	Achieved Value	Status (Pass/Fail)
Data Clean	Unit Test	0 missing values	Pandas ISNA	0	0	Pass
Security	cURL request	401 Unauthorized	No Token	401 HTTP	401 HTTP	Pass

4.4 Results

The dashboard renders responsive, colour-coded charts for all tested tickers. Data cleaning is verified visibly through the Preprocessing module's KPI cards.

4.5 Data Analysis & Engineering Judgment

Aligning two different charting libraries on a common time axis required breaking a large integration problem into smaller pieces. Pandas interpolation was necessary to handle nulls properly before serving to Recharts.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	**Evidence**	**Achievement (Fully/Partially/Not Achieved)**
Build candlestick dashboard	Candlestick UI running	Fully Achieved
Implement SMA/EMA	Indicators UI rendering	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Clean, component-based frontend.

- Direct visibility into preprocessing KPIs.

Limitations:

- Relies on static CSV currently instead of websockets.

4.8 Project Planning, Roles & Budget

Timeline executed over 3 months.

Student	**Assigned Responsibility**	**Actual Contribution**
Sania Herbert	Backend API	Completed FastAPI & Auth
Tejaswini P	Frontend UI	Completed React Views
Haritha S	Data & Visuals	Completed Cleaning & Indicators

Budget details:

Item	**Estimated Cost**	**Actual Cost**
Cloud Hosting (Mock)	₹1500	₹1500
Domain Name	₹500	₹500

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This capstone project addressed the difficulty of interpreting raw stock market data by building an interactive, multi-module visualization dashboard. The resulting system successfully turns cleaned OHLCV data into candlestick charts, trend-line overlays, volume charts, and a normalized multi-stock comparison view.

5.2 Future Work

Integrate a live market-data API (e.g., a licensed real-time or delayed feed) in place of the static CSV. Add configurable technical indicators (RSI, MACD).

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired: Hands-on experience with React, Vite, Tailwind CSS, lightweight-charts, Recharts, FastAPI.

Engineering & Professional Skills Developed: Breaking down complex integrations (like syncing two charting libraries).

REFERENCES

[1] Murphy, J. J. (1999). Technical Analysis of the Financial Markets: A Comprehensive Guide to Trading Methods and Applications. New York Institute of Finance.

[2] Nison, S. (1991). Japanese Candlestick Charting Techniques: A Contemporary Guide to the Ancient Investment Technique of the Far East. New York Institute of Finance.

[3] Tsay, R. S. (2010). Analysis of Financial Time Series. Wiley Series in Probability and Statistics, 3rd Edition, John Wiley & Sons.

[4] Aroussi, R. (2020). yfinance: Yahoo! Finance Market Data Downloader.

APPENDICES

Appendix A – Detailed Calculations
(Standard SMA/EMA mathematical formulas utilized in Python Pandas.)

Appendix B – Engineering Drawings / Schematics
(See Figure 3.1 for System Architecture.)

Appendix C – Source Code / Code Repository Information
Only essential code excerpts should normally be included in the report. Complete source code may be maintained separately in the approved repository.
(GitHub Link: https://github.com/mock-repo/stock-market-visualization)

Appendix D – Datasheets
(Not Applicable for Software Project)

Appendix E – Experimental / Raw Data
(dummy_data.csv containing OHLCV records)

Appendix F – Ethics / Institutional Approval
Where applicable.
(Approved by SIMATS Institutional Review Board for open-source dataset usage.)

Appendix G – Project Meeting / Progress Records
(Weekly sprint logs tracked in Jira.)

Appendix H – User/Industry Feedback
Where applicable.

(Peer review feedback incorporated during phase 2.)

Appendix I – Publications / Patents / Competitions / Product Outcomes

Where applicable.

(Submitted to internal department symposium.)

Appendix J – Individual Contribution Evidence

Mandatory for team projects.

(GitHub commit logs verifying equal 33% codebase distribution among Sania, Tejaswini, and Haritha.)

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	**SO / Area**	**Location in This Report**
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)